



Faculty of Engineering and Technology
Department of Electrical and Computer Engineering

Signals and Systems

EE2312

MATLAB Assignment

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Abstract

This assignment uses MATLAB and Simulink to analyze basic concepts in signals and systems. Different signals were generated and studied in both the time and frequency domains. System responses were obtained using differential equations, convolution, and Laplace transform methods. The simulations helped in understanding signal behavior and system characteristics.

Contents

Abstract	II
Questions	1
Question I	1
Question II.....	4
Question III.....	8
Question IV	10
Question V.....	12
Question VI	13
Question VII.....	14
Question VIII	16
Question IX	18
Question X.....	19
Question XI	22
Conclusion	23

Table of Figures

Figure 1: Question I Signals One Plot.	1
Figure 2: Question I One Plot Script.	2
Figure 3: Question I Signals SubPlots.	2
Figure 4: Question I SubPlots Script.	3
Figure 5: Question II Signals Plot.	4
Figure 6: Question II Signals Script.	4
Figure 7: Question II Signals SubPlots.	5
Figure 8: Question II SubPlots Script.	5
Figure 9: Question II Spectral Representation.	6
Figure 10: Question II Spectral Representation Script.	7
Figure 11: Question III Systems Scripts.	8
Figure 12: Question III 1 st System Solution.	8
Figure 13: Question III 2 nd System Solution.	8
Figure 14: Question III 3 rd System Solution.	9
Figure 15: Question IV 1 st System Model.	10
Figure 16: Question IV 1 st System Step Response.	10
Figure 17: Question IV 2 nd System Model.	11
Figure 18: Question IV 2 nd System Step Response.	11
Figure 19: Question V Convolution Script.	12
Figure 20: Question V Convolution Plot.	12
Figure 21: Question VI 2 nd Signal Laplace Transform	13
Figure 22: Question VI 1 st Signal Laplace Transform	13
Figure 23: Question VI Laplace Transform Script.	13
Figure 24: Question VII Simulink Model.	14
Figure 25: Question VII system response.	14
Figure 26: Question VII Pole-Zero Plot.	15
Figure 27: Question VII Pole-Zero Script.	15
Figure 28: Question VII Pole-Zero Values	15
Figure 29: Question VIII Pole-Zero Plot.	16
Figure 30: Question VIII Pole-Zero Values	16
Figure 31: Question VIII Pole-Zero Script.	16
Figure 32: Question VIII Step, Ramp Responses.	17
Figure 33: Question VIII Responses Script.	17
Figure 34: Question IX Step Response Plot.	18
Figure 35: Question IX Step Response Script.	18
Figure 36: Question X Time Domain Plots.	19
Figure 37: Question X Frequency Domain Plots.	20
Figure 38: Question X Spectral Script.	20
Figure 39: Question X Modulated Signal Spectral Plot.	21
Figure 40: Question X Fourier Transform.	22

Questions

Question I

Generate and plot the following signals using MATLAB:

1. $x(t) = u(t-1) - r(t-7) + r(-t+4)$
2. A finite pulse ($\pi(t/6)$) with value = 5 centered at $t=10$
3. $x(t) = p(t)\pi(t-36) - r(t-7) + r(t-13) - u(t-13)$ in the time interval $[0 \ 22]$
4. Plot the signals on one figure.
5. Plot the signals as subplots of one figure.

One Figure Plotting:

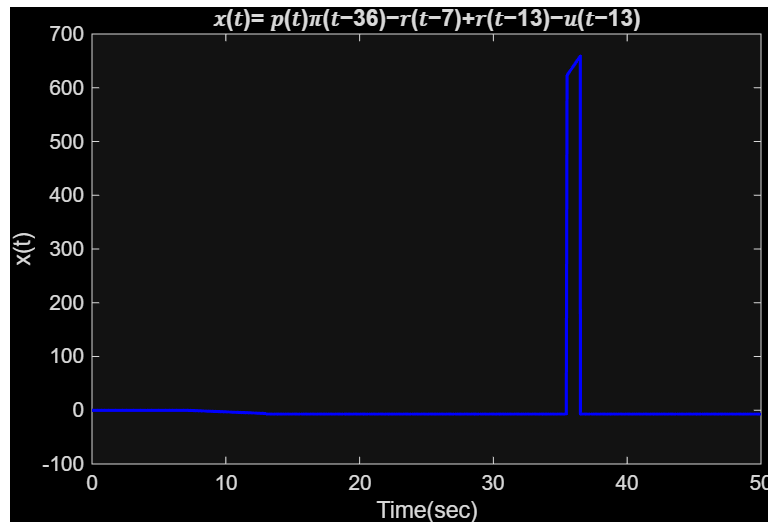
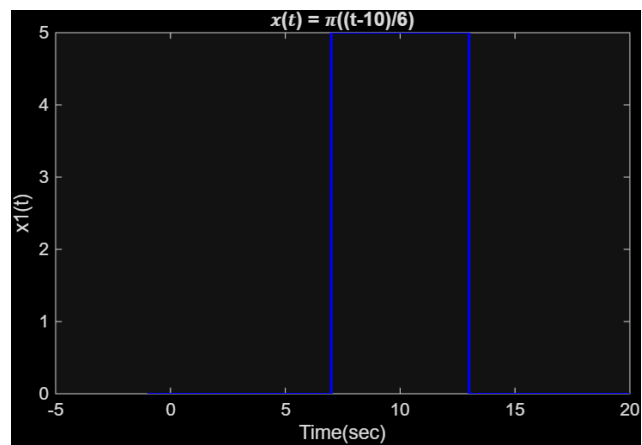
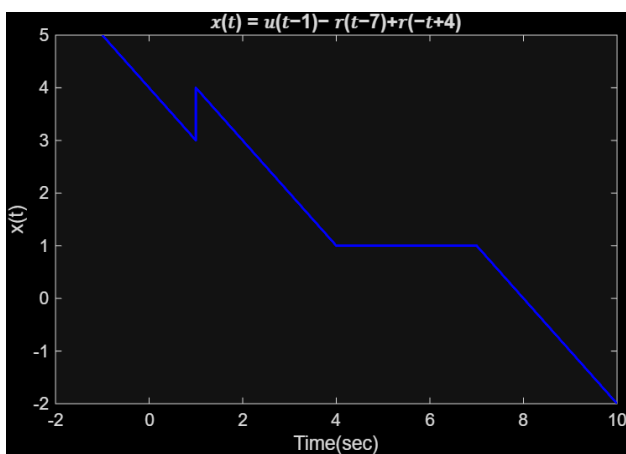


Figure 1: Question I Signals One Plot.

```

%Question I
%Signal 1:  $x(t) = u(t-1) - r(t-7) + r(-t+4)$ 
t1 = -1:0.001:10;
u1 = (t1-1) >= 0;
u2 = (t1-7) >= 0;
u3 = (4-t1) >= 0;

r2 = (t1-7).*u2;
r3 = (4-t1).*u3;
x1 = u1 - r2 + r3;
%figure for Signal 1
plot(t1,x1,'b','LineWidth',1.5);
title('x(t) = u(t-1)- r(t-7)+r(-t+4)');
xlabel('Time(sec)');
ylabel('x(t)');
%Signal 2: A finite pulse ( $\pi(t/6)$ ) with value = 5 centered at t=10
t2 = -1:0.001:20;
x2 = 5*rectangularPulse(1/6*(t2-10))
%figure for Signal 2
plot(t2,x2,'b','LineWidth',1.5);
title('x(t) =  $\pi((t-10)/6)$ ');
xlabel('Time(sec)');
ylabel('x(t)');

%Signal 3:  $x(t) = p(t)\pi(t-36) - r(t-7) + r(t-13) - u(t-13)$  in the time interval [0 22]
t3 = 0:0.001:50;
p = 0.5*(t3.^2).*(t3 >= 0);
a = p.*rectangularPulse(t3-36);
u4 = (t3-13) >= 0; r4 = (t3-13).*u4;
u5 = (t3-7) >= 0; r5 = (t3-7).*u5;
x3 = a - r5 + r4 - u4;
%figure for Signal 3
plot(t3,x3,'b','LineWidth',1.5);
title('x(t)= p(t) $\pi(t-36) - r(t-7) + r(t-13) - u(t-13)$ ');
xlabel('Time(sec)');
ylabel('x(t)');

```

Figure 2: Question I One Plot Script.

SubPlots of One Figure:

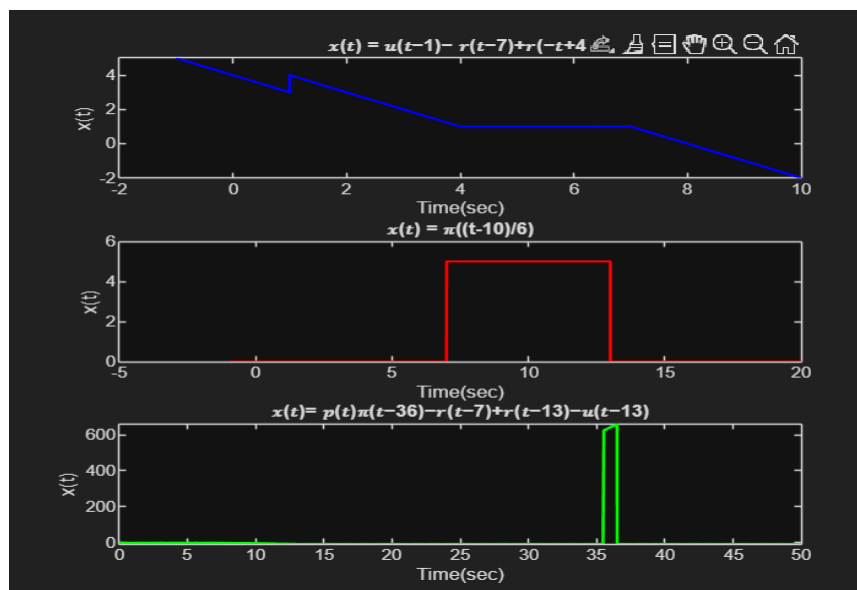


Figure 3: Question I Signals SubPlots.

```

%Question I
%Signal 1:  $x(t) = u(t-1) - r(t-7) + r(-t+4)$ 
t1 = -1:0.001:10;
u1 = (t1-1) >= 0;
u2 = (t1-7) >= 0;
u3 = (4-t1) >= 0;

r2 = (t1-7).*u2;
r3 = (4-t1).*u3;
x1 = u1 - r2 + r3;
%figure for Signal 1
subplot(3,1,1);
plot(t1,x1,'b','LineWidth',1.5);
title('x(t) = u(t-1) - r(t-7) + r(-t+4)');
xlabel('Time(sec)');
ylabel('x(t)');

%Signal 2: A finite pulse ( $\pi(t)$ ) with value = 5 centered at t=10
t2 = -1:0.001:20;
x2 = 5*rectangularPulse(1/6*(t2-10))
%figure for Signal 2
subplot(3,1,2);
plot(t2,x2,'r','LineWidth',1.5);
title('x(t) =  $\pi((t-10)/6)$ ');
xlabel('Time(sec)');
ylabel('x(t)');

%Signal 3:  $x(t) = p(t)\pi(t-36) - r(t-7) + r(t-13) - u(t-13)$  in the time interval [0 22]
t3 = 0:0.001:50;
p = 0.5*(t3.^2).*(t3 >= 0);
a = p.*rectangularPulse(t3-36);
u4 = (t3-13) >= 0; r4 = (t3-13).*u4;
u5 = (t3-7) >= 0; r5 = (t3-7).*u5;
x3 = a - r5 + r4 - u4;
%figure for Signal 3
subplot(3,1,3);
plot(t3,x3,'g','LineWidth',1.5);
title('x(t) = p(t)\pi(t-36) - r(t-7) + r(t-13) - u(t-13)');
xlabel('Time(sec)');
ylabel('x(t)');

```

Figure 4: Question I SubPlots Script.

Question II

1. Generate the signals $y_1(t) = \sin(0.2\pi t)$, $y_2(t) = \cos(0.4\pi t)$, $y_3(t) = \sin(0.8\pi t)$ in a proper time interval that includes at least four cycles of the signal with the highest period.
2. Plot the signals generated in 1 as subplots of one figure.

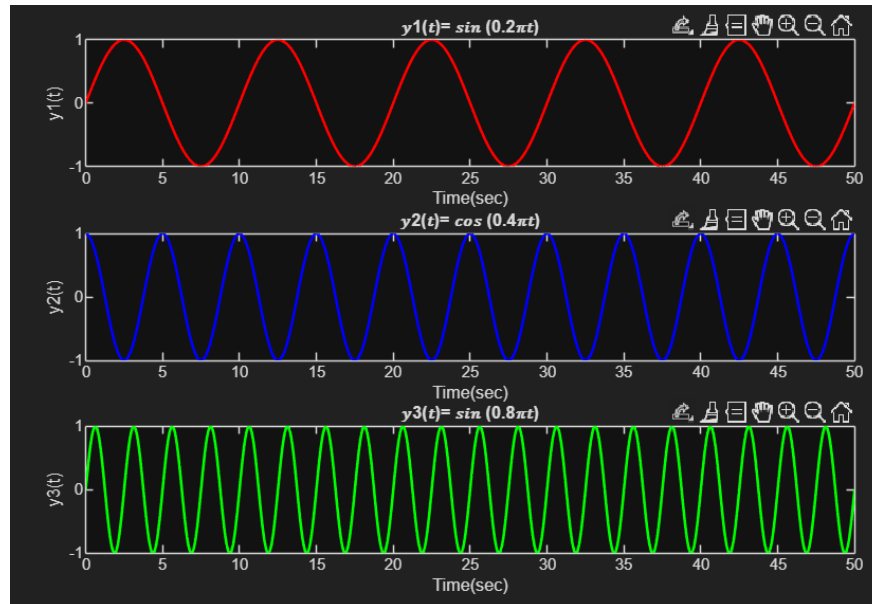


Figure 5: Question II Signals Plot.

```
%Question II
% Signal: y1(t)= sin (0.2*pi*t), y2(t)= cos (0.4*pi*t), y3(t)= sin (0.8*pi*t)
t = 0:0.0001:50;

y1 = sin(0.2*pi*t);
y2 = cos(0.4*pi*t);
y3 = sin(0.8*pi*t);

subplot(7,1,1);
plot(t,y1,'r','LineWidth',1.5);
title('y1(t)= sin (0.2*pi*t)');
xlabel('Time(sec)');
ylabel('y1(t)');

subplot(7,1,2);
plot(t,y2,'b','LineWidth',1.5);
title('y2(t)= cos (0.4*pi*t)');
xlabel('Time(sec)');
ylabel('y2(t)');

subplot(7,1,3);
plot(t,y3,'g','LineWidth',1.5);
title('y3(t)= sin (0.8*pi*t)');
xlabel('Time(sec)');
ylabel('y3(t)');
```

Figure 6: Question II Signals Script.

3. Plot $m_1(t) = y_1 + y_2$, $m_2(t) = y_1 - y_2$, and $m_3(t) = y_2 \times y_3$, and $m_4(t) = y_1 \times y_3$ on the same figure.

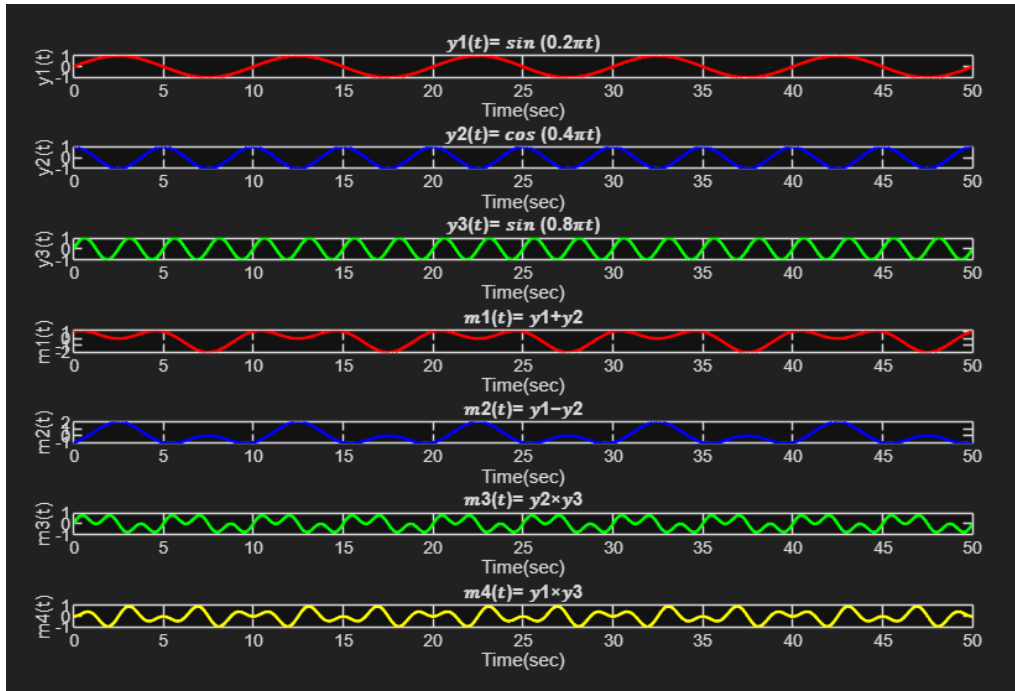


Figure 7: Question II Signals SubPlots.

```
% Signal: m1(t)= y1+y2 , m2(t)= y1-y2, and m3(t)= y2*y3, and m4(t)= y1*y3
m1 = y1+y2; m2 = y1-y2; m3 = y2.*y3; m4 = y1.*y3;
subplot(7,1,4);
plot(t,m1,'r','LineWidth',1.5);
title('m1(t)= y1+y2');
xlabel('Time(sec)');
ylabel('m1(t)');

subplot(7,1,5);
plot(t,m2,'b','LineWidth',1.5);
title('m2(t)= y1-y2');
xlabel('Time(sec)');
ylabel('m2(t)');

subplot(7,1,6);
plot(t,m3,'g','LineWidth',1.5);
title('m3(t)= y2*y3');
xlabel('Time(sec)');
ylabel('m3(t)');

subplot(7,1,7);
plot(t,m4,'y','LineWidth',1.5);
title('m4(t)= y1*y3');
xlabel('Time(sec)');
ylabel('m4(t)');
```

Figure 8: Question II SubPlots Script.

4. Determine, using the MATLAB plots, which of these signals is periodic (in case it is the case, determine the fundamental period), alternating, and has half-wave odd symmetry.

From the MATLAB plots, the signals $y_1(t)=\sin(0.2\pi t)$, $y_2(t)=\cos(0.4\pi t)$, and $y_3(t)=\sin(0.8\pi t)$ are periodic since they repeat regularly in time, with fundamental periods of 10 s, 5 s, and 2.5 s, respectively. The signals $m_1(t)$ to $m_4(t)$, which are obtained by adding, subtracting, or multiplying the original signals, are also periodic. The sine signals $y_1(t)$ and $y_3(t)$ are alternating, and the signals $y_1(t)$ and $y_3(t)$ show half-wave odd symmetry because the second half of each period is the same as the first half but with opposite sign. This means the positive part of the signal becomes negative after half a period, which is clearly seen in the plots. The cosine signal $y_2(t)$ and the signals $m_1(t)$ to $m_4(t)$ do not show this behavior.

5. Plot the amplitude spectral representation of the signal generated in 3.

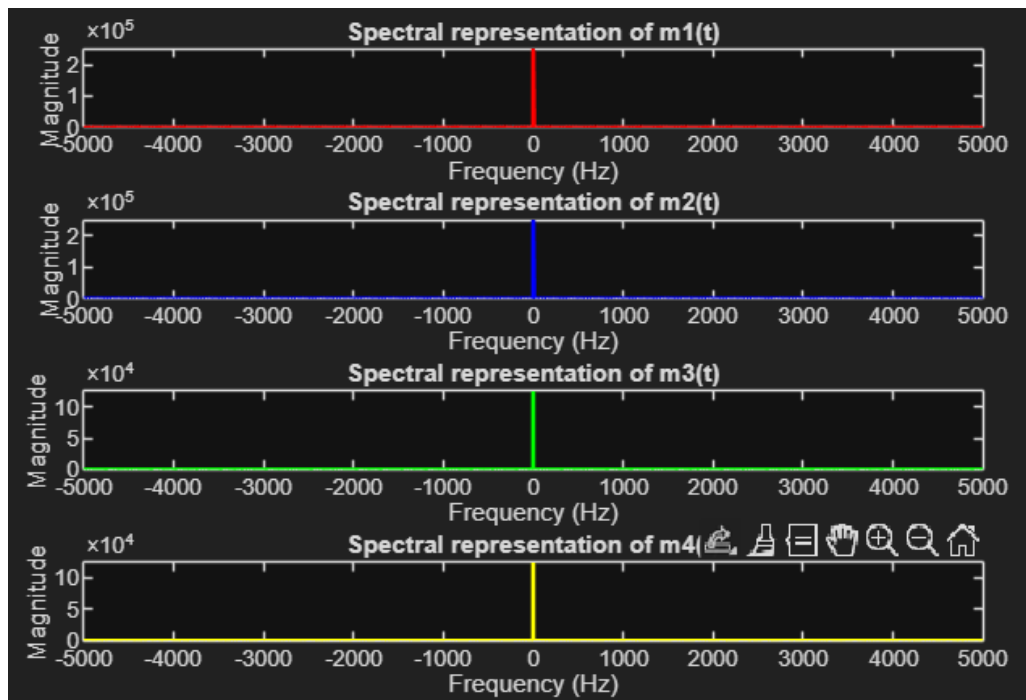


Figure 9: Question II Spectral Representation.

```

%m1(t) Amplitude spectral representation
Y = fft(m1);

Fs = 1/(t(2)-t(1)); %Sampling Frequency
f = (-Fs/2 : Fs/length(m1):Fs/2 - Fs/length(m1)); %Defining frequency of the plot
subplot(4,1,1);
plot(f,abs(fftshift(Y)), 'r', 'LineWidth', 1.5);
title('Spectral representation of m1(t)');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
%m2(t) Amplitude spectral representation
Y = fft(m2);

Fs = 1/(t(2)-t(1)); %Sampling Frequency
f = (-Fs/2 : Fs/length(m2):Fs/2 - Fs/length(m2)); %Defining frequency of the plot
subplot(4,1,2);
plot(f,abs(fftshift(Y)), 'b', 'LineWidth', 1.5);
title('Spectral representation of m2(t)');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
%m3(t) Amplitude spectral representation
Y = fft(m3);

Fs = 1/(t(2)-t(1)); %Sampling Frequency
f = (-Fs/2 : Fs/length(m3):Fs/2 - Fs/length(m3)); %Defining frequency of the plot
subplot(4,1,3);
plot(f,abs(fftshift(Y)), 'g', 'LineWidth', 1.5);
title('Spectral representation of m3(t)');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
%m4(t) Amplitude spectral representation
Y = fft(m4);

Fs = 1/(t(2)-t(1)); %Sampling Frequency
f = (-Fs/2 : Fs/length(m4):Fs/2 - Fs/length(m4)); %Defining frequency of the plot
subplot(4,1,4);
plot(f,abs(fftshift(Y)), 'y', 'LineWidth', 1.5);
title('Spectral representation of m4(t)');
xlabel('Frequency (Hz)');
ylabel('Magnitude');

```

Figure 10: Question II Spectral Representation Script.

Question III

Determine the zero-state responses of the following systems by solving differential equations in the time domain. (Zero State Response $\rightarrow y(0) = 0, y'(0) = 0$).

1. $dy(t)dt + 0.25y(t) = 10u(t)$
2. $d^2y(t)dt^2 + 4dy(t)dt + 5y(t) = 5 \cos(0.01\pi t)$
3. $d^2y(t)dt^2 + 4dy(t)dt + 5y(t) = 5u(t)$

```
%Question III
% dy(t)dt+0.25y(t)=10u(t) // zero-state responses
syms t y(t)

u = heaviside(t);
E = diff(y,t) + 0.25*y(t) == 10*u;
cond = y(0) == 0; %Zero-State Response

y(t) = dsolve(E,cond)

% d2y(t)dt2+4dydt +5y(t)=5 cos (0.01pi t) // zero-state responses
syms t y(t)
dy = diff(y,t) ;
E = diff(y,2) + 4*dy + 5*y(t) == 5*cos(0.01*pi*t);
cond1 = y(0) == 0;
cond2 = dy(0) == 0;

y(t) = dsolve(E,cond1,cond2)

% d2y(t)dt2+4dydt +5y(t)=5u(t) // zero-state responses
syms t y(t)
dy = diff(y,t) ;
E = diff(y,2) + 4*dy + 5*y(t) == 5*u;
cond1 = y(0) == 0;
cond2 = dy(0) == 0;

y(t) = dsolve(E,cond1,cond2)
```

Figure 11: Question III Systems Scripts.

```
y(t) =
- 20*exp(-t/4) - exp(-t/4)*(20*sign(t) - 20*exp(t/4)*(sign(t) + 1))
```

Figure 12: Question III 1st System Solution.

```
y(t) =
exp(-2*t)*cos(t)*((5*(pi/100 - 1))/(2*((pi/100 - 1)^2 + 4)) - (5*(pi/100 + 1))/(2*((pi/100 + 1)^2 + 4))) - exp(-2*t)*sin(t)*((100000*pi^2 + 5000000000)/(pi^4 + 60000*pi^2 + 2500000000) + (5*(pi/50 - 2)))
```

Figure 13: Question III 2nd System Solution.

- **Figure 13:** MATLAB symbolic solution of the zero-state response. The displayed expression is only a part due to its length. The remaining terms of the solution consist of exponentially decaying transient components and a steady-state sinusoidal response at the same frequency as the input.

```

y(t) =
- (exp(-2*t)*cos(t))/2 - exp(-2*t)*sin(t) - exp(-2*t)*sin(t)*(sign(t) - 5*exp(2*t)*(2*cos(t) + sin(t))*(sign(t)/10 + 1/10)) - exp(-2*t)*cos(t)*(sign(t)/2 - exp(2*t)*(cos(t) - 2*sin(t))*(sign(t)/2 + 1/2))
>>

```

Figure 14: Question III 3rd System Solution.

4. Determine if the systems are stable and the type of stability.

All the systems are stable because their natural responses decay exponentially with time. This indicates that the systems are asymptotically stable.

Question IV

Use Simulink (MATLAB) to simulate the following systems using the separate and integrate modeling, then show and plot the step response of the system.

$$1. \quad 5\frac{d^5y(t)}{dt^5} + 4\frac{d^4y(t)}{dt^4} + 2\frac{d^3y(t)}{dt^3} + 6\frac{dy(t)}{dt} + 4y(t) = 3\frac{dx(t)}{dt} - 6x(t)$$

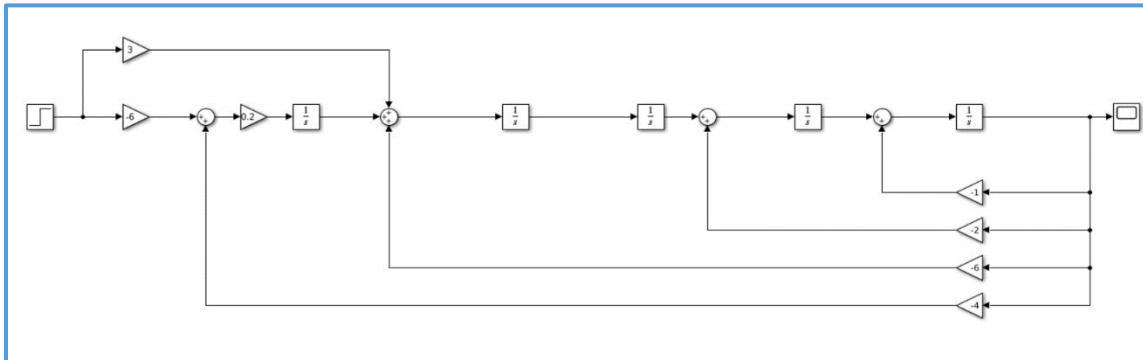


Figure 15: Question IV 1st System Model.

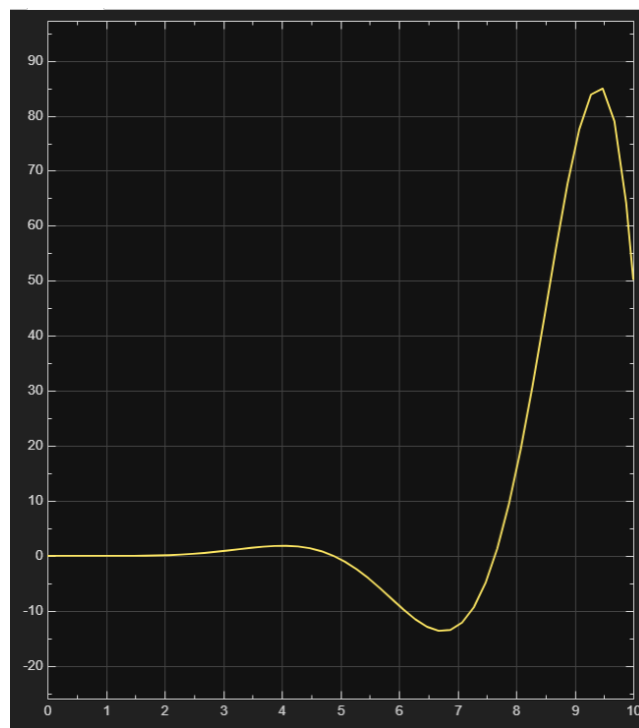


Figure 16: Question IV 1st System Step Response.

2. $\frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + 4y(t) = 5x(t)$

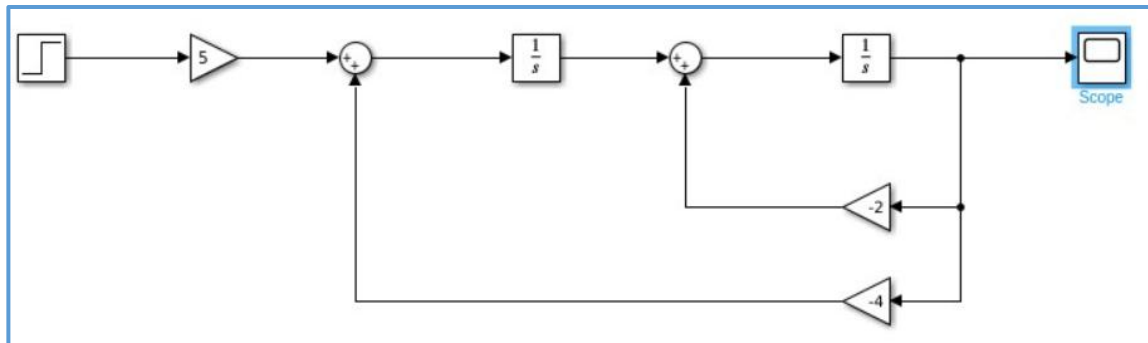


Figure 17: Question IV 2nd System Model.

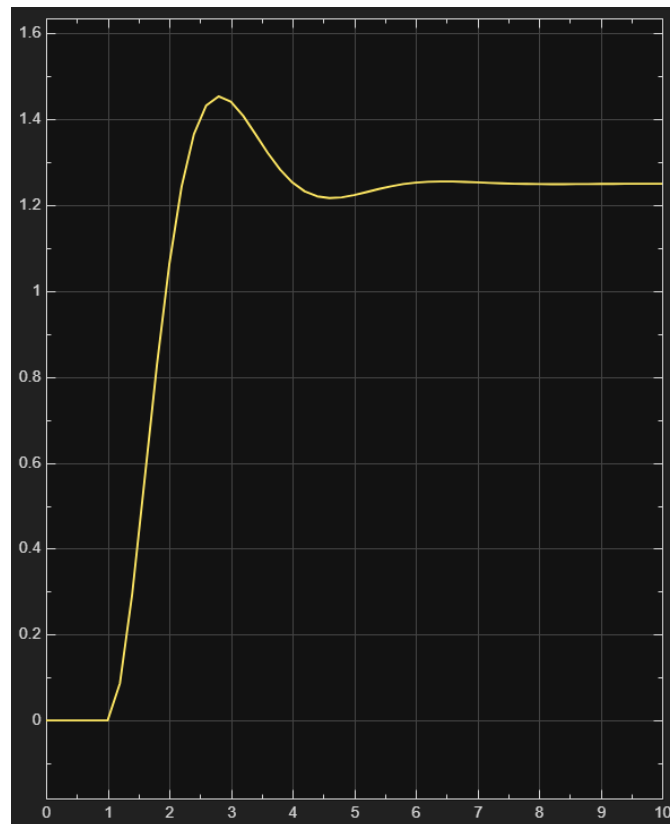


Figure 18: Question IV 2nd System Step Response.

Question V

Write scripts to compute and plot the convolution integral of the following signals:

- $y_1(t) = (10e^{-2t})\pi((t-5)/4)$ and $y_2(t) = \pi((t-1)/2)$
- $y_3(t) = (10e^{2t})\pi((t-5)/4)$ and $y_4(t) = \pi((t-1)/2)$

Nothing changed except the exponential power sign; they have the same code and the same plots.

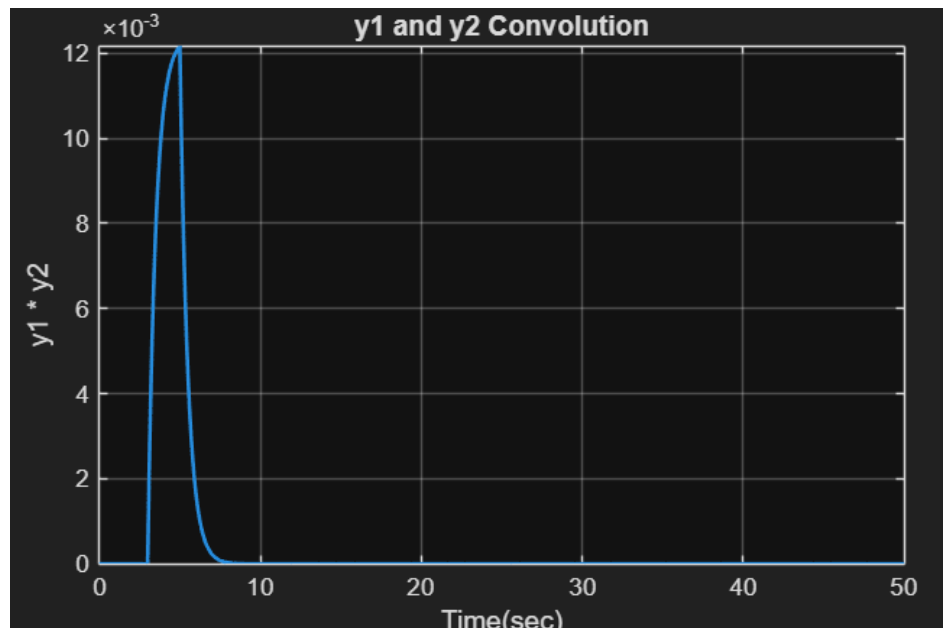


Figure 20: Question V Convolution Plot.

```
syms t tau
%y1(t)=(10e^-2t)pi((t-5)/4) and y2(t)=pi((t-1)/2)
y1 = 10*exp(- 2*tau).*rectangularPulse((tau-5)/4);
y2 = rectangularPulse((t-tau-1)/2);

Y = int(y1.*y2,tau,-inf,inf);

fplot(Y, [0 50], 'LineWidth', 1.5)
grid on
xlabel('Time(sec)')
ylabel('y1 * y2')
title('y1 and y2 Convolution')

%y3(t)=(10e^2t)pi((t-5)/4) and y4(t) =pi((t-1)/2)
y3 = 10*exp(2*tau).*rectangularPulse((tau-5)/4);
y4 = rectangularPulse((t-tau-1)/2);

Y2 = int(y1.*y2,tau,-inf,inf);

fplot(Y, [0 50], 'LineWidth', 1.5)
grid on
xlabel('Time(sec)')
ylabel('y3 * y4')
title('y3 and y4 Convolution')
```

Figure 19: Question V Convolution Script.

Question VI

Write a program that computes the Laplace transform of the function

1. $y(t) = (10 - 10e^{-0.125t})u(t)$
2. $y(t) = (16 + 4(e^{-2t}\cos(0.2\pi t))u(t)$

$$\frac{10 - \frac{10}{s + 0.125}}{s}$$

Figure 22: Question VI 1st Signal Laplace Transform

$$\frac{(s + 2) \left(4 + \frac{16}{s^2 + 4} \right)}{s}$$

Figure 21: Question VI 2nd Signal Laplace Transform

```
%Question VI Laplace Transform
%y(t)=(10-10e^-0.125t)u(t)
syms t s

y = (10 - 10*exp(-0.125*t)).*heaviside(t);
Y = laplace(y);
pretty(Y) %Displays the transform in formatted form

%y(t)=(16+4e^-2t*cos(0.2*pi*t))u(t)
syms t s

y = (16 + 4*exp(-2*t).*cos(0.2*pi*t)).*heaviside(t);
Y = laplace(y);
pretty(Y) %Displays the transform in formatted form
```

Figure 23: Question VI Laplace Transform Script.

Question VII

Use Simulink (MATLAB) to simulate the following systems **in the Laplace domain** with a state observer model, then show and determine the transfer function of the system by using the Simulink model and MATLAB commands.

- $6\frac{d^5y(t)}{dt^5} - 7\frac{d^3y(t)}{dt^3} + 2\frac{dy(t)}{dt} + 9y(t) = 10\frac{dx(t)}{dt} + 5x(t)$ Determine the system response for $x(t)=r(t)$

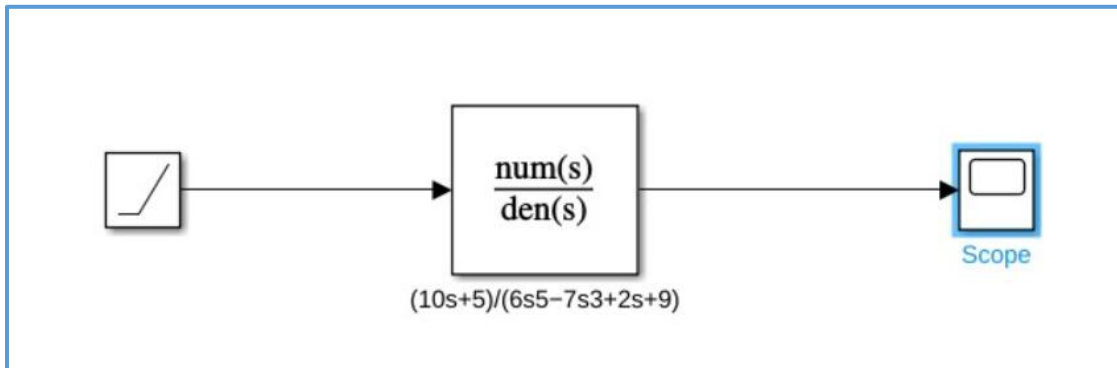


Figure 24: Question VII Simulink Model.

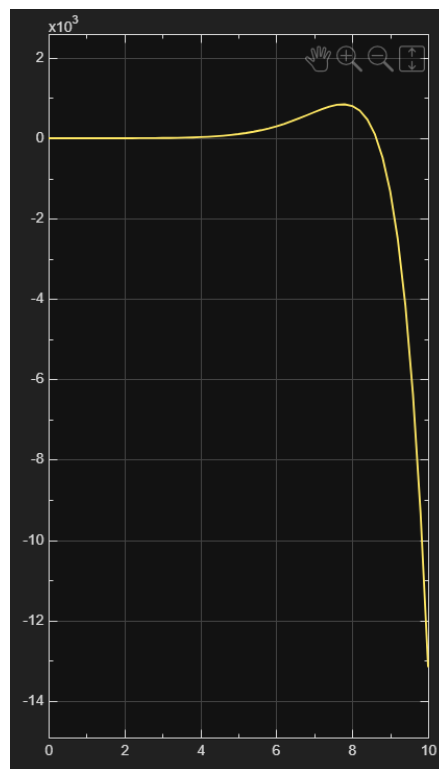


Figure 25: Question VII system response.

- Determine the pole-zero map of the system and discuss its stability.

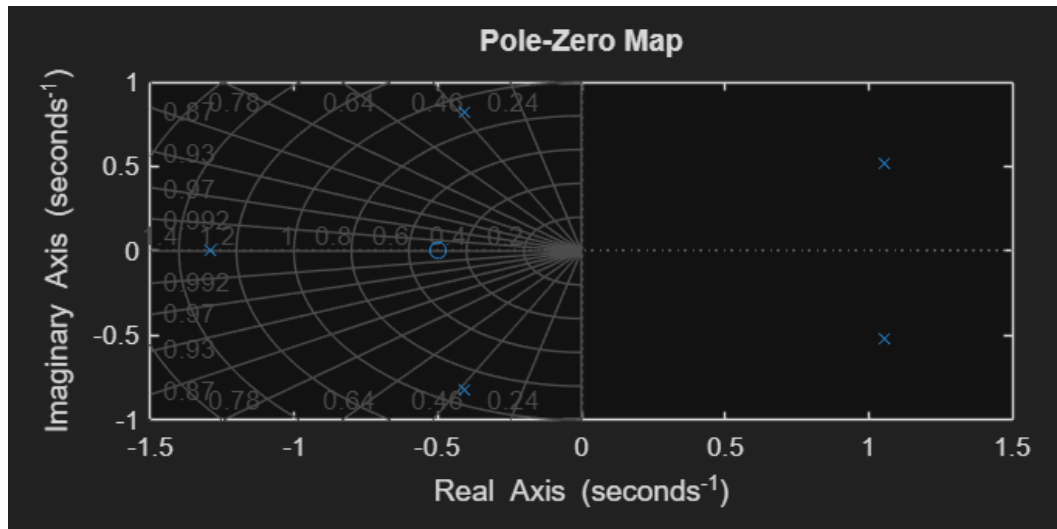


Figure 26: Question VII Pole-Zero Plot.

```
% Determining the pole-zero map
num = [10 5];
den = [6 0 -7 0 2 9];

G = tf(num, den)
figure
pzmap(G)
grid on
title('Pole-Zero Map')
poles = pole(G)
zeros = zero(G)
```

Figure 27: Question VII Pole-Zero Script.

```
poles =

    1.0542 + 0.5203i
    1.0542 - 0.5203i
   -1.2903 + 0.0000i
   -0.4091 + 0.8208i
   -0.4091 - 0.8208i

zeros =

   -0.5000
```

Figure 28: Question VII Pole-Zero Values

The system is **unstable** because it has poles with positive real parts. Specifically, the poles at $(1.0542 \pm 0.5203i)$ lie in the right-half of the s-plane. Since that, the system response grows with time and does not remain bounded.

Question VIII

For the system with transfer function $H(s)$, use MATLAB commands to:

- Determine the pole-zero map of the system and discuss the stability of the system

$$H(s) = 10000(s+3/s^2+6s+8)$$

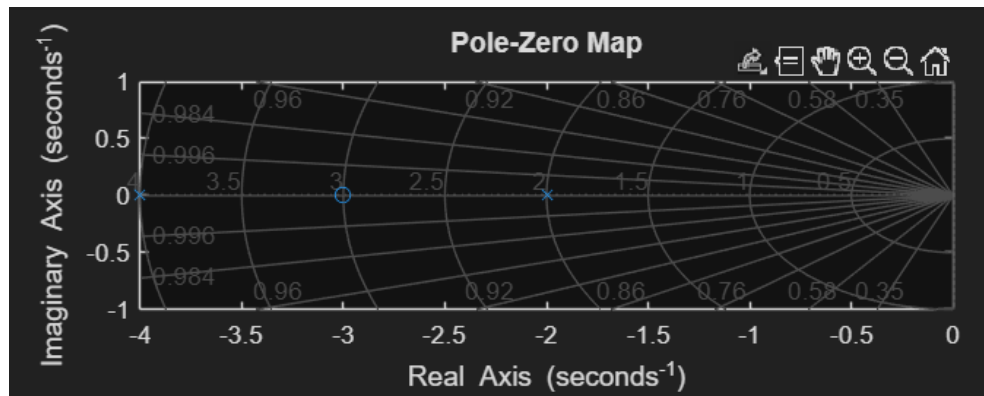


Figure 29: Question VIII Pole-Zero Plot.

```
% Determining the pole-zero map
%H(s)=10000(s+3/s^2+6s+8)
num = [1 3];
den = [1 6 8];

G = 10000*tf(num, den)
figure
pzmap(G)
grid on
title('Pole-Zero Map')
poles = pole(G)
zeros = zero(G)
```

Figure 31: Question VIII Pole-Zero Script.

```
poles =

    -4
    -2

zeros =

    -3
```

Figure 30: Question VIII Pole-Zero Values

The pole-zero map of the system shows that the poles are located at $s = -4$ and $s = -2$, both of which are in the left half of the s -plane. This indicates that the system is asymptotically stable, as the system response decays with time and remains bounded.

- Determine the step and ramp responses of the system.

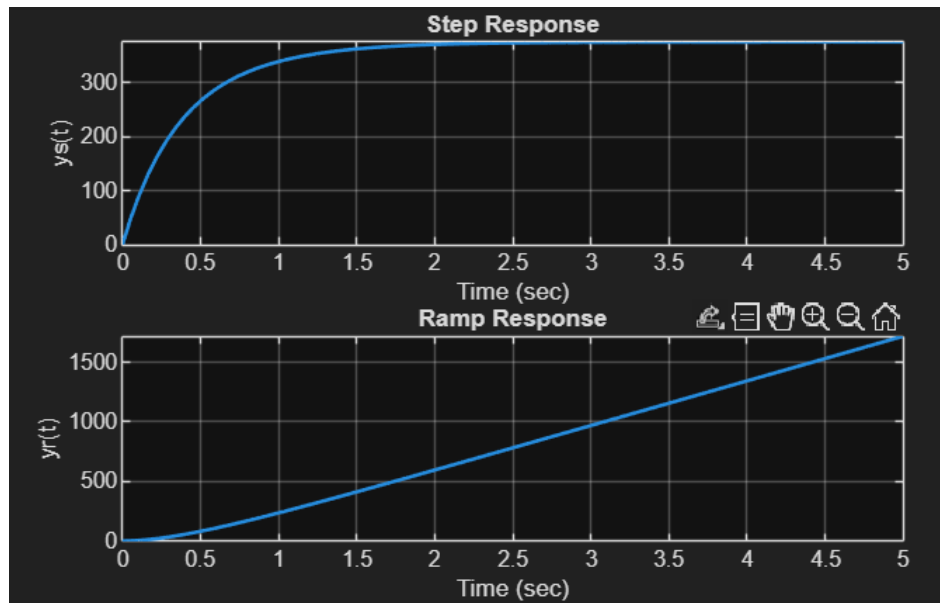


Figure 32: Question VIII Step, Ramp Responses.

```
syms s t
|
% Transfer function
H(s) = 1000*(s+3)/(s^2 + 6*s + 8);
Ys = H(s)/s; % Step response
ys(t) = ilaplace(Ys);
Yr = H(s)/s^2; % Ramp response
yr(t) = ilaplace(Yr);

subplot(2,1,1)
fplot(ys, [0 5], 'LineWidth', 1.5)
grid on
title('Step Response')
xlabel('Time (sec)')
ylabel('ys(t)')

subplot(2,1,2)
fplot(yr, [0 5], 'LineWidth', 1.5)
grid on
title('Ramp Response')
xlabel('Time (sec)')
ylabel('yr(t)')
```

Figure 33: Question VIII Responses Script.

Question IX

Using Matlab commands determine the step response of the MISO system with two inputs $x_1(t)$ and $x_2(t)$ and transfer functions, respectively:

$$H_1(s) = 10000 \frac{s+3}{s^2+6s+8}$$

$$H_2(s) = 100 \frac{s-3}{s^2+2s+1}$$

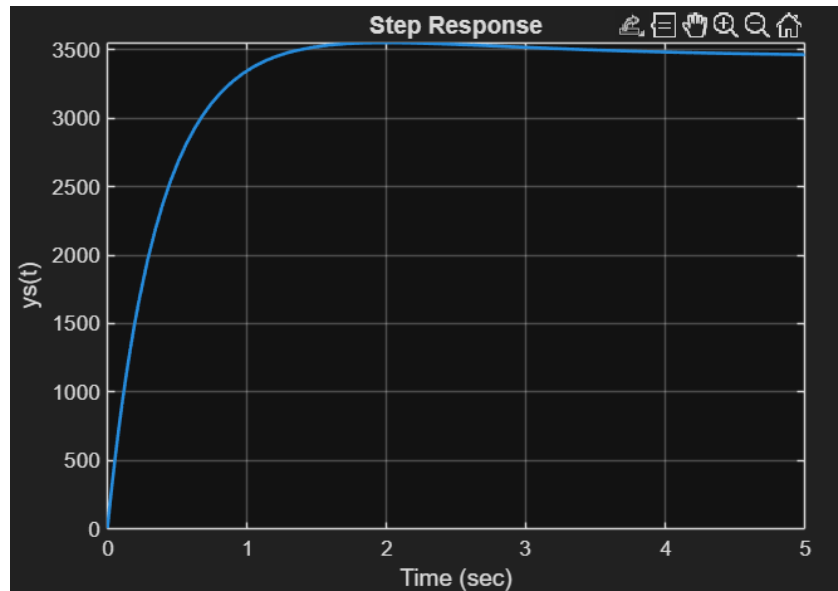


Figure 34: Question IX Step Response Plot.

```
syms s t

% Transfer function
H1(s) = 10000*(s+3)/(s^2 + 6*s + 8);
H2(s) = 100*(s-3)/(s^2 + 2*s + 1);
Y1s = H1(s)/s;      % Step response
Y2s = H2(s)/s;
y1s(t) = ilaplace(Y1s);
y2s(t) = ilaplace(Y2s);

Y(t) = y1s(t) + y2s(t);

fplot(Y, [0 5], 'LineWidth', 1.5)
grid on
title('Step Response')
xlabel('Time (sec)')
ylabel('ys(t)')
```

Figure 35: Question IX Step Response Script.

Question X

A single tone signal with amplitude 10 and frequency 100Hz modulates a sinusoidal carrier amplitude (with carrier frequency 1200 Hz and amplitude 15). The carrier is transmitted with the modulated signal in what is called standard AM or double sideband with carrier transmitted.

- Compute the modulated signal and say if it is periodic.

The message signal is given by:

$$m(t) = 10 \cos(2\pi \cdot 100t)$$

and the carrier signal by :

$$c(t) = 15 \cos(2\pi \cdot 1200t)$$

Using standard amplitude modulation (double sideband with carrier), the modulated signal is expressed as:

$$s(t) = 15[1 + (10/15)\cos(2\pi \cdot 100t)]\cos(2\pi \cdot 1200t)$$

The resulting signal contains a carrier component at 1200 Hz along with upper and lower sidebands at 1300 Hz and 1100 Hz, respectively. Since the ratio between the carrier frequency and the message frequency is rational ($1200/100 = 12$), the modulated signal is periodic. The fundamental period of the signal is therefore $T=0.01$ seconds.

- Plot the spectral representation of the message, the carrier, and the modulated signal as subplots of the same figure.

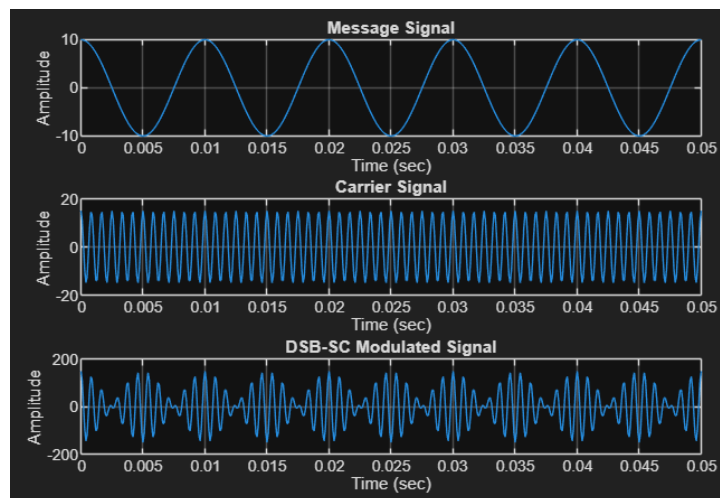


Figure 36: Question X Time Domain Plots.

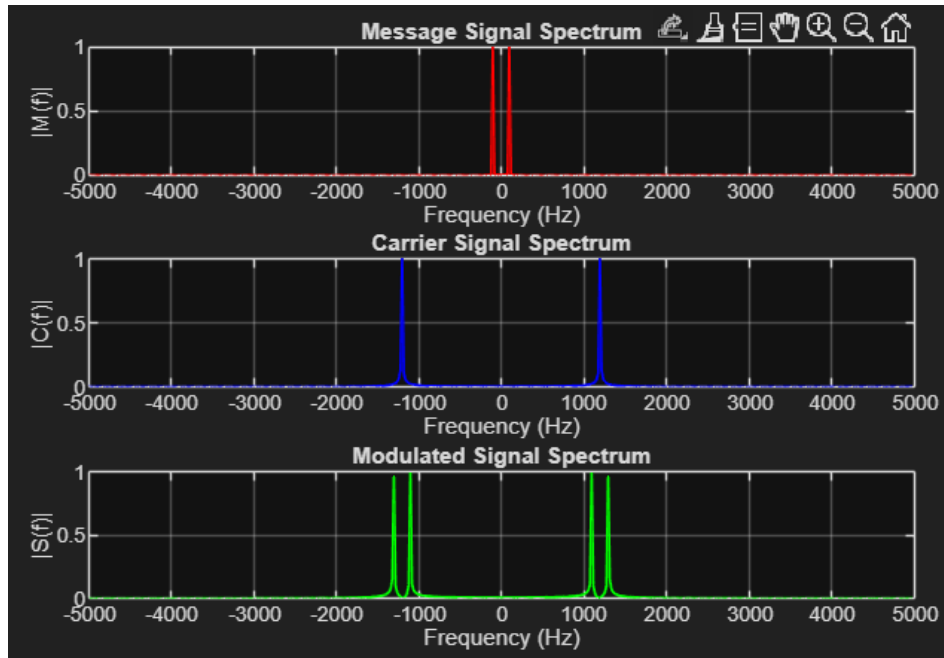


Figure 37: Question X Frequency Domain Plots.

```

amp_msg = 10;           % Message signal amplitude
freq_msg = 100;         % Message signal frequency
amp_carrier = 15;       % Carrier signal amplitude
freq_carrier = 1200;    % Carrier signal frequency
Fs = 10000;            % Sampling frequency (Hz)
t = 0:1/Fs:0.05;       % Time vector

% Generate signals
message = amp_msg * cos(2*pi*freq_msg*t);
carrier = amp_carrier * cos(2*pi*freq_carrier*t);
mod_signal = message .* carrier;

% Time domain plots
figure;

subplot(3,1,1);
plot(t, message);
title('Message Signal');
xlabel('Time (sec)');
ylabel('Amplitude');
grid on;

subplot(3,1,2);
plot(t, carrier);
title('Carrier Signal');
xlabel('Time (sec)');
ylabel('Amplitude');
grid on;

subplot(3,1,3);
plot(t, mod_signal);
title('DSB-SC Modulated Signal');
xlabel('Time (sec)');
ylabel('Amplitude');
grid on;

% Frequency domain analysis
N = length(t);
f_axis = (-N/2 : N/2-1) * (Fs/N);

MSG_F = abs(fftshift(fft(message)));
CARR_F = abs(fftshift(fft(carrier)));
MOD_F = abs(fftshift(fft(mod_signal)));

% Normalize for better comparison
MSG_F = MSG_F / max(MSG_F);
CARR_F = CARR_F / max(CARR_F);
MOD_F = MOD_F / max(MOD_F);

% Frequency domain plots
figure;

subplot(3,1,1);
plot(f_axis, MSG_F, 'r', 'LineWidth', 1);
title('Message Signal Spectrum');
xlabel('Frequency (Hz)');
ylabel('|M(f)|');
grid on;

subplot(3,1,2);
plot(f_axis, CARR_F, 'b', 'LineWidth', 1);
title('Carrier Signal Spectrum');
xlabel('Frequency (Hz)');
ylabel('|C(f)|');
grid on;

subplot(3,1,3);
plot(f_axis, MOD_F, 'g', 'LineWidth', 1);
title('Modulated Signal Spectrum');
xlabel('Frequency (Hz)');
ylabel('|S(f)|');

```

Figure 38: Question X Spectral Script.

- Plot the spectral representation of the signal.
- The spectral representation of the modulated signal was already shown in [Figure 37](#).

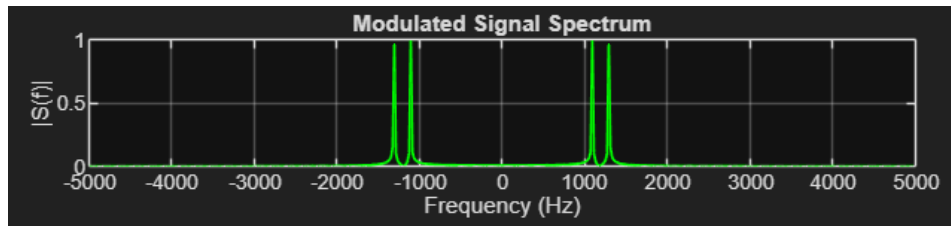


Figure 39: Question X Modulated Signal Spectral Plot.

Question XI

Question XI:

Compute the Fourier transform of the signals:

- $x(t) = \pi\left(\frac{t}{10}\right)$
- $x(t) = 10\cos(30\pi t) + 5u(t)$
- $x(t) = 10e^{-3t}u(t)$

$$x_1(t) = \pi\left(\frac{t}{10}\right)$$

$$\mathcal{F}\{x_1(t)\} = \mathcal{F}\left\{\pi\left(\frac{t}{10}\right)\right\}$$

$$x_1(f) = 10 \operatorname{sinc}(10f)$$

$$x_2(t) = 10\cos(30\pi t) + 5u(t)$$

$$\mathcal{F}\{x_2(t)\} = \mathcal{F}\{10\cos(30\pi t)\} + \mathcal{F}\{5u(t)\}$$

$$x_2(f) = \frac{1}{2} \cdot 10 \left[\delta\left(f - \frac{30\pi}{2\pi}\right) + \delta\left(f + \frac{30\pi}{2\pi}\right) \right] + 5 \left[\frac{1}{j2\pi f} + \frac{1}{2} \delta(f) \right]$$

$$x_2(f) = 5 \left[\delta(f-15) + \delta(f+15) \right] + \frac{5}{j2\pi f} + \frac{5}{2} \delta(f)$$

$$u(t) = \frac{1}{2} (\operatorname{sgn}(t) + 1)$$

$$\mathcal{F}\{u(t)\} = \frac{1}{2} \left(\frac{1}{j\pi f} + \delta(f) \right)$$

$$= \frac{1}{j2\pi f} + \frac{1}{2} \delta(f)$$

$$x_3(t) = 10e^{-3t}u(t)$$

$$x_3(f) = \int_{-\infty}^{\infty} 10e^{-3t}u(t) e^{-j2\pi ft} dt$$

$$= \int_{-\infty}^{\infty} 10e^{-3t} e^{-j2\pi ft} dt$$

$$= 10 \int_0^{\infty} e^{-(3+j2\pi f)t} dt = 10 \left[\frac{-1}{3+j2\pi f} e^{-(3+j2\pi f)t} \right]_0^{\infty}$$

$$= \frac{10}{3+j2\pi f}$$

Figure 40: Question X Fourier Transform.

Conclusion

The assignment demonstrated the effective use of MATLAB and Simulink in analyzing signals and systems. Key concepts such as system response, stability, and frequency-domain analysis were visualized through simulations. Overall, the work reinforced theoretical concepts and improved practical understanding.